

Mining Social Networks

COMPSCI 753

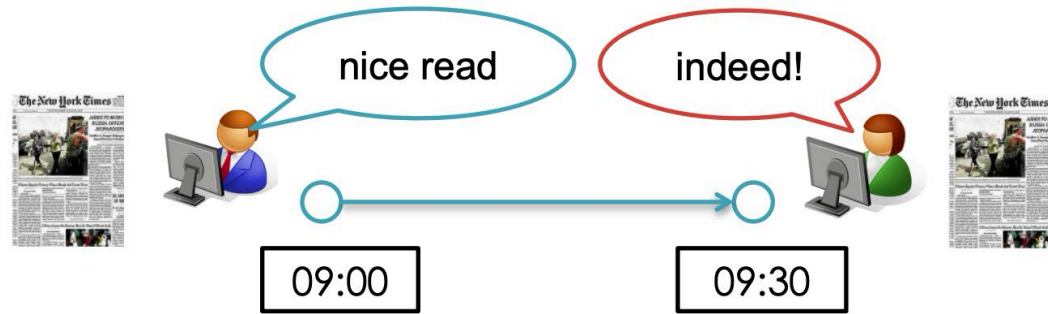
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Overview

- Social network analysis
 - Social network introduction
 - Community detection
 - Influence maximization

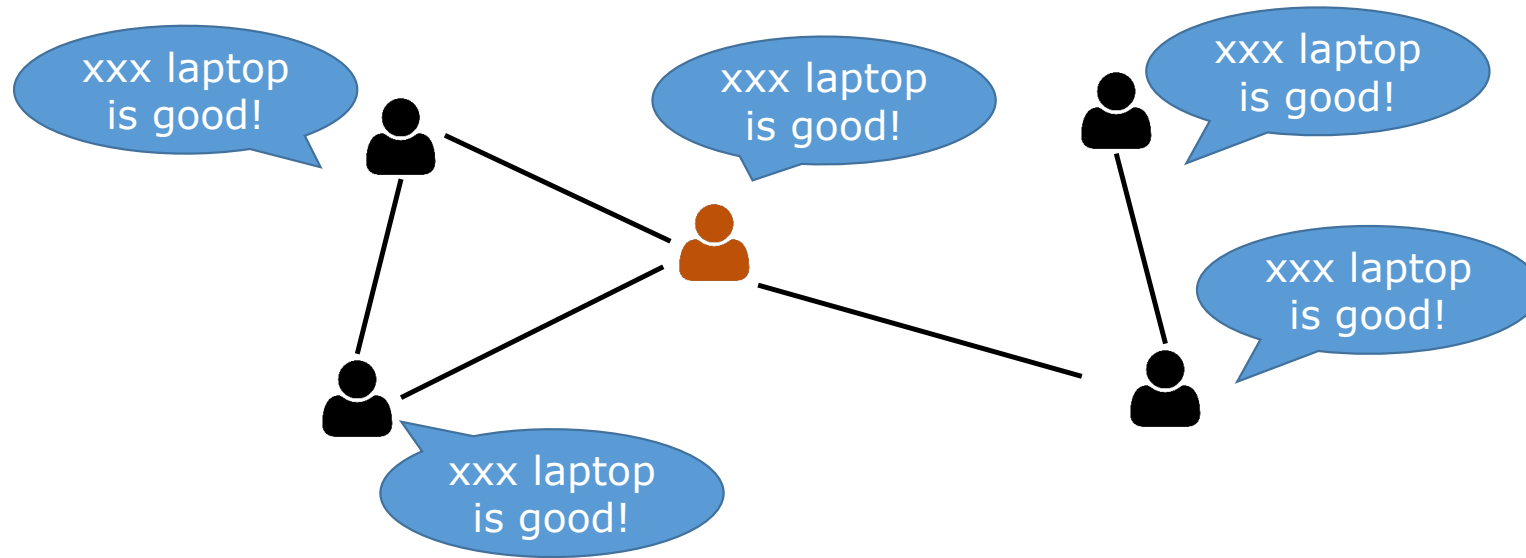
Influence

- People are connected and perform actions



Application: viral marketing

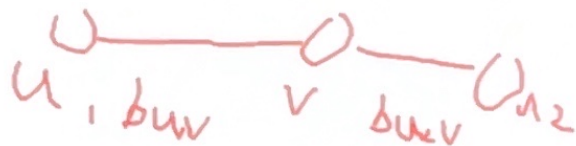
- Idea: exploit social influence for marketing
- Basic assumption: word-of-mouth effect
- More efficient marketing strategy – target the influencers



Problem definition

- Given a social network $G = (V, E)$, find a seed-set S of k influential people in V such that by targeting them we maximize the **inference spread**.
- Influence spread: Given a set $S \subseteq V$, the influence spread $\sigma(S)$ of S is the expected number of influenced users after the propagation terminates if the users in S were selected as seed users
- Propagation models:
 - Independent Cascade Model
 - Linear Threshold Model

Linear Threshold (LT) Model



- Each edge $(u, v) \in E$ has weight b_{uv} , **where** $\sum_{\{u | u \rightarrow v \in E\}} b_{uv} \leq 1$ i.e. the sum of weights of all u nodes connecting to v is up to 1

- Let S_t be the set of active nodes at time t . Initially, only seed nodes in S are activated at time $t = 0$, i.e., $S_0 = S$

- Randomly** pick an activation threshold $\theta_u \sim U[0, 1]$ for each node u

- At each time step $t > 0$, a node u gets activated if:

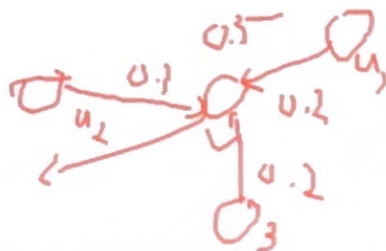
- The sum of weights from all its activated in-neighbor, i.e.,

$$\sum_{\{w | w \rightarrow u \in E \text{ \& } w \in S_{t-1}\}} b_{wu} \geq \theta_u.$$

$$0.7 > 0.4$$

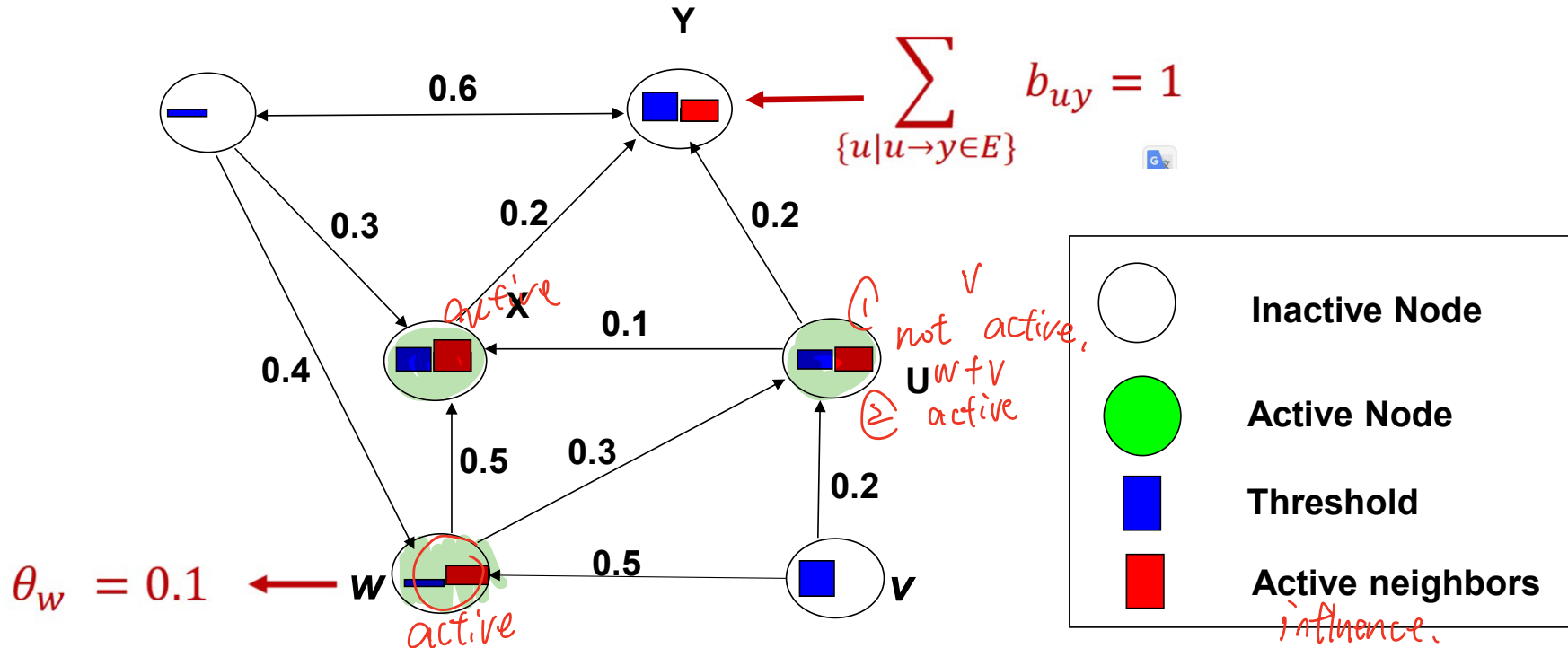
active the node.

0.6



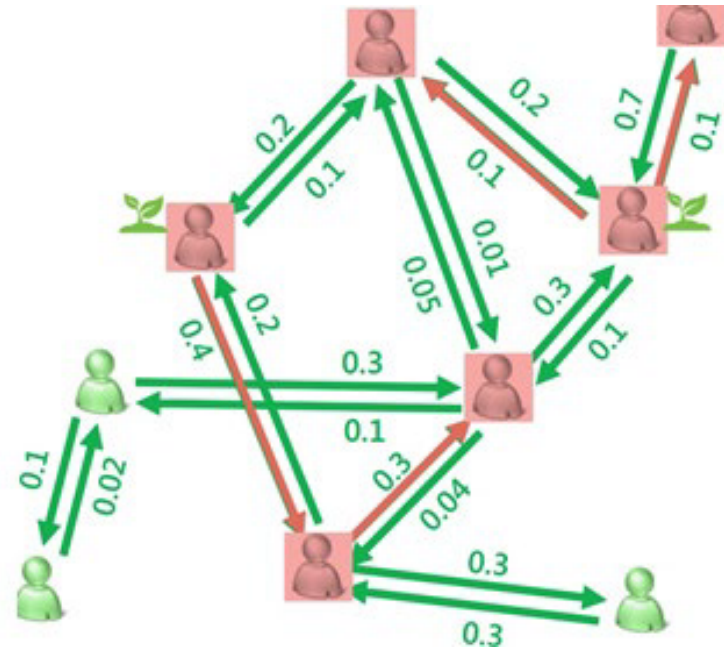
$$0.2 + 0.3 + 0.2 = 0.7$$

Example – Linear Threshold

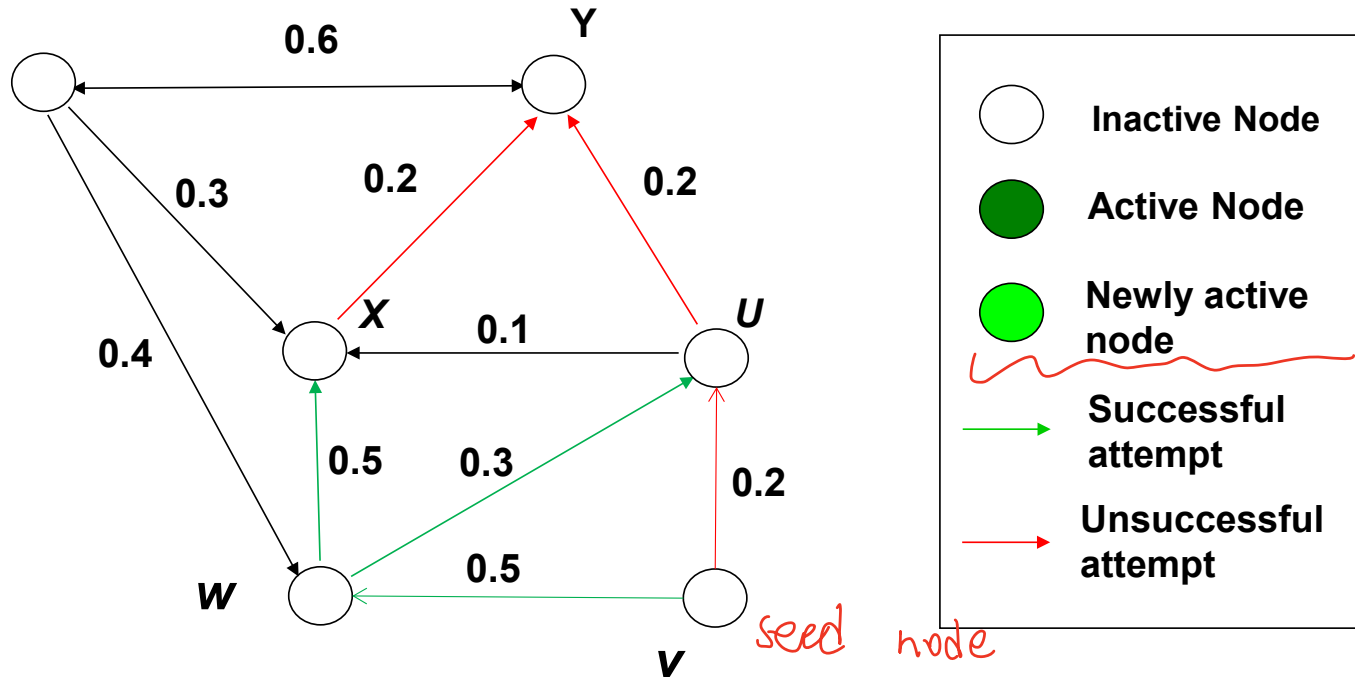


Independent Cascade (IC)

- Each edge $(u, v) \in E$ has influence probability p_{uv} .
- Seeds S are activated at time $t = 0$, e.g., $S_0 = S$
- At each time step $t > 0$, each active node u **independently** activates its inactive neighbor v :
 - Succeed with probability p_{uv} and fails with probability $1 - p_{uv}$.
 - Once a node is active, it cannot become inactive

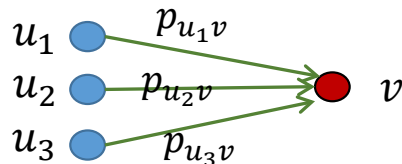


Example – Independent Cascade

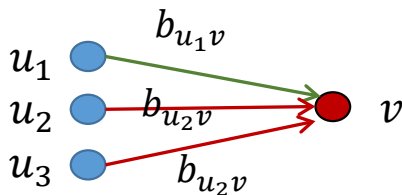


Mutually-exclusive Cascade (MC)

- In IC, a node is activated independently by its in-neighbors



- In mutually-exclusive cascade model (MC), only one in-neighbors of v activates v



Question:

What is the probability that v is activated by u_1 , but not by other nodes in these two models?

IC model:

$$\begin{aligned} & \text{Prob}[v \text{ is activated by } u_1] \\ & \times \text{Prob}[v \text{ is NOT activated by } u_2] \\ & \times \text{Prob}[v \text{ is NOT activated by } u_3] \\ & = p_{u_1v} \times (1 - p_{u_2v}) \times (1 - p_{u_3v}) \end{aligned}$$

MC model:

b_{u_1v} - Prob[v is activated by u_1 , but not others]

What is the probability that v is activated by any of its in-neighbor?

$$P_{IC}(v) = 1 - \prod_{i=1}^n (1 - p_{u_i v})$$

$$P_{MC}(v) = \sum_{i=1}^n b_{u_i v}$$

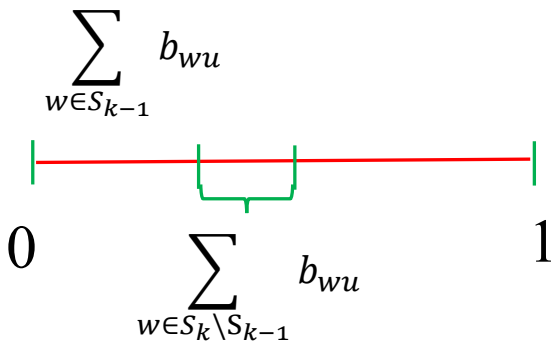
LT = MC

Linear Threshold

Prob[u is not activated in step k ,
but is activated at step $k + 1]$

$$= \frac{\sum_{w \in S_k \setminus S_{k-1}} b_{wu}}{1 - \sum_{w \in S_{k-1}} b_{wu}}$$

即之前未激活的节点
在后续步骤中激活的影响

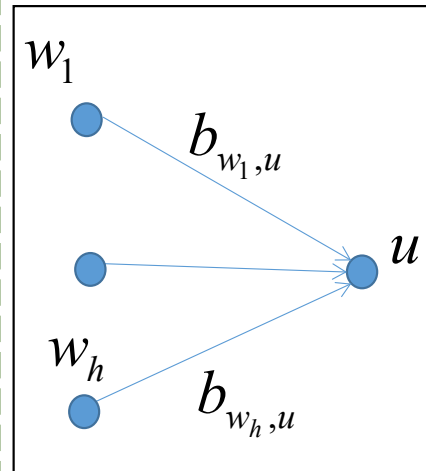


Mutually-exclusive Cascade

Prob[u is not activated in step k , but
is activated at step $k + 1] =$

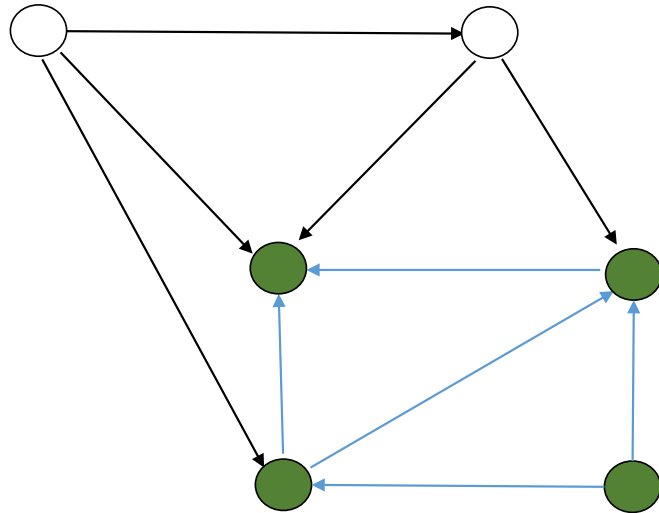
$$\frac{\text{Prob}[u \text{ is activated by } w \text{ in } S_k \setminus S_{k-1}]}{\text{Prob}[u \text{ is not activated by } w \text{ in } S_{k-1}]}$$

$$= \frac{\sum_{w \in S_k \setminus S_{k-1}} b_{wu}}{1 - \sum_{w \in S_{k-1}} b_{wu}}$$



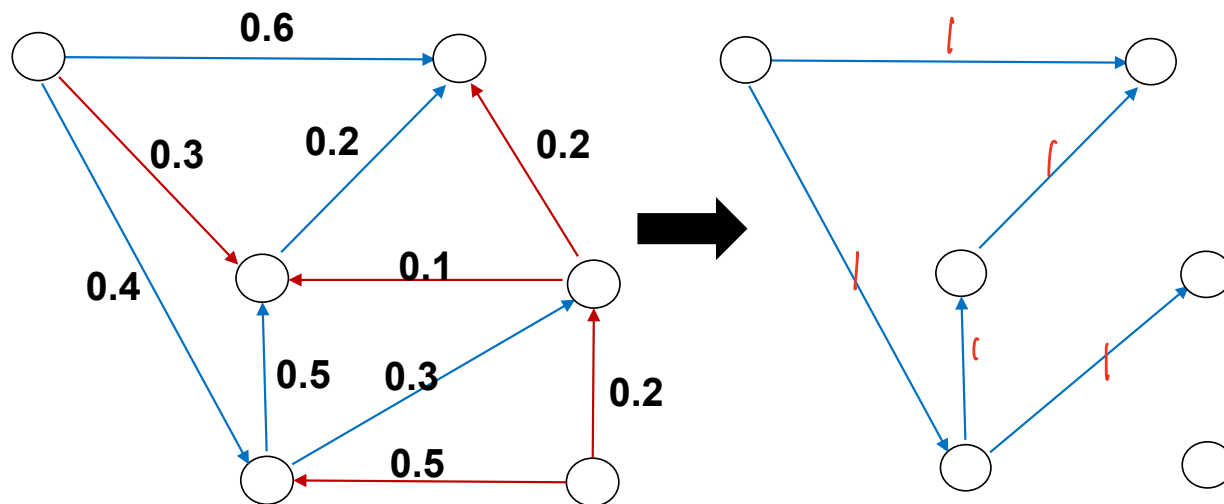
Computing Influence Spread

- Deterministic graph
 - Accessibility from the active nodes



Computing Influence Spread

- **How to compute the influence of IC and MC (or LT)?**
 - Stochastic model is annoying! → deterministic models?
 - Enumerate all possible sets of active edges - activated edges are *live edges*
 - Let X be the set denoting the label of *edges (live or blocked)*



$X_{uv} = 1$ if edge $u \rightarrow v$ is live

$$\text{Prob}[X] = \prod_{(u,v) \in E} p_{uv}^{X_{uv}} (1 - p_{uv})^{1-X_{uv}}$$

Computing Influence Spread

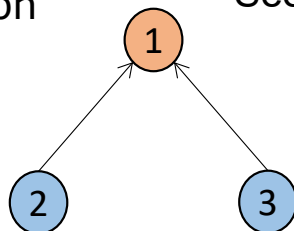
- **How to compute the influence of IC and MC (or LT)?**
 1. Calculate each possible deterministic graph with edge label X and its probability $\text{prob}[X]$
 2. Compute the number of nodes that can be accessed from the seed set S in the deterministic graph $\sigma^X(S)$
 3. The influence spread of the seed set is the expectation over all the possible edge labels X , $\sigma(S) = \sum_X \text{prob}[X] \cdot \sigma^X(S)$

Example – IC and MC

$z^2 = \varphi$ 12

IC

Graph



Seed set {2,3}

IC

$$p(AB) = p_{21}p_{31}$$

$$p(A\bar{B}) = p_{21}(1 - p_{31})$$

$$p(\bar{A}B) = (1 - p_{21})p_{31}$$

$$p(\bar{A}\bar{B}) = (1 - p_{21})(1 - p_{31})$$

	active	
	A (2 → 1)	$\bar{A}$ (2 ↗ 1)
B (3 → 1)		
$\bar{B}$ (3 ↗ 1)		

能达到的，激活的节点数

$$\sigma_{IC}(\{2,3\}) = E[\sigma_{IC}^X(\{2,3\})] = \sum_X p(X) \cdot \sigma_{IC}^X(\{2,3\})$$

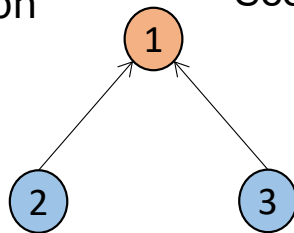
$$= 3 \cdot p(AB) + 3 \cdot p(A\bar{B}) + 3 \cdot p(\bar{A}B) + 2 \cdot p(\bar{A}\bar{B})$$

$$= 2 + p_{21} + p_{31} - p_{21}p_{31}$$

Example – IC and MC

MC.

Graph



Seed set {2,3}

MC

$$p(AB) = 0$$

$$p(A\bar{B}) = b_{21}$$

$$p(\bar{A}B) = b_{31}$$

$$p(\bar{A}\bar{B}) = 1 - b_{21} - b_{31}$$

只允许一个节点激活。

	$A (2 \rightarrow 1)$	$\bar{A} (2 \nrightarrow 1)$
$B (3 \rightarrow 1)$		
$\bar{B} (3 \nrightarrow 1)$		

$$\begin{aligned} \sigma_{MC}(\{2,3\}) &= E[\sigma_{MC}^X(\{2,3\})] = \sum_X p(X) \cdot \sigma_{MC}^X(\{2,3\}) \\ &= 3 \cdot p(AB) + 3 \cdot p(A\bar{B}) + 3 \cdot p(\bar{A}B) + 2 \cdot p(\bar{A}\bar{B}) \\ &= 2 + b_{21} + b_{31} \end{aligned}$$

Summary

- Influence Maximization
 - Finding a set of influential nodes in the network
 - Two propagation models
 - Linear Threshold *influence*
 - Independent Cascade *possibility*
 $\{N_c\}$

Summary for the Graph Algorithms

- Web search – Fighting with spams
 - PageRank – dead-ends and spider traps
 - Biased PageRank – topic-specific, personalized pagerank
 - Link spam and link farm
- Community detection – Finding people with strong relationships.
 - Edge betweenness
 - Graph cut
 - Modularity
- Influence Maximization – Finding influential users in the network
 - Two propagation models: LT and IC